with $X, Y \neq 0$. But $F_i^+ V$ is injective at i, since the maps are canonical projections, whose direct sum is the tautological embedding. Therefore X and Y also have to be injective at i and hence (by Proposition 6.6.6)

$$F_i^+ F_i^- X = X, \quad F_i^+ F_i^- Y = Y.$$

In particular

$$F_i^- X \neq 0, \quad F_i^- Y \neq 0.$$

Therefore

$$V = F_i^- F_i^+ V = F_i^- X \oplus F_i^- Y,$$

which is a contradiction, since V was assumed to be indecomposable. So we can infer that

$$F_i^+ V$$

is indecomposable. □

Proposition 6.6.8. *Let Q be a quiver and let V be a representation of Q.*

(1) *Let $i \in Q$ be a sink and let V be surjective at i. Then*

$$d(F_i^+ V) = s_i(d(V)).$$

(2) *Let $i \in Q$ be a source and let V be injective at i. Then*

$$d(F_i^- V) = s_i(d(V)).$$

Proof. We only prove the first statement; the second one follows similarly. Let $i \in Q$ be a sink and let

$$\varphi : \bigoplus_{j \to i} V_j \to V_i$$

be surjective. Let $K = \ker \varphi$. Then

$$\dim K = \sum_{j \to i} \dim V_j - \dim V_i.$$

Therefore we get

$$\big(d(F_i^+ V) - d(V)\big)_i = \sum_{j \to i} \dim V_j - 2 \dim V_i = -B\left(d(V), \alpha_i\right)$$

and

$$\big(d(F_i^+ V) - d(V)\big)_j = 0, \quad j \neq i.$$